

## Brainstorming & Idea Prioritization Template

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 27 june 2026                    |
| <b>Team ID</b>       | XXXXXX                          |
| <b>Project Name</b>  | Credit Card Approval Prediction |
| <b>Maximum Marks</b> | 3 Marks                         |

### Step 1: Brainstorm and Idea Listing

Each team member lists out as many ideas as possible without judging them at this stage.

| S.No | Team Member | Idea / Suggestion                                                                                                                                                     | Category                  | Group No. |
|------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------|
| 1    | Member 1    | Use an automated script to consolidate multi-table datasets (application_record and credit_record) based on unique customer IDs.                                      | Data Integration          | Group 1   |
| 2    | Member 2    | Build a binary classification targeting function that flags a customer as "High Risk" if they are more than 30 days overdue in any given month.                       | Target Engineering        | Group 2   |
| 3    | Member 1    | Handle massive class imbalance (overwhelming number of approved status records) by applying structural algorithmic penalization options like class_weight='balanced'. | Data Imbalance Mitigation | Group 3   |
| 4    | Member 3    | Train and benchmark multiple predictive architectures (Logistic Regression, Decision                                                                                  | Model Benchmarking        | Group 4   |

|   |          |                                                                                                                                                                           |                         |         |
|---|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------|
|   |          | Trees, and Random Forests) using Scikit-Learn to select the most robust engine.                                                                                           |                         |         |
| 5 | Member 2 | Handle the <b>30.5% missing values</b> in the OCCUPATION_TYPE attribute by applying structural categorical string imputation rather than deleting entries.                | Missing Data Imputation | Group 1 |
| 6 | Member 3 | Convert text attributes into numerical arrays using LabelEncoder, serialize the optimal trained classifier via pickle, and build a web application interface using Flask. | Model Deployment        | Group 5 |

### Step 2: Idea Prioritization

Rate each grouped idea on feasibility and importance, then select the final idea(s) to move forward with.

| Group No. | Final Idea                                                                                                                                                                                  | Feasibility (High/Medium/Low) | Importance (High/Medium/Low) | Priority | Selected (Yes/No) |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------------------------------|----------|-------------------|
| 1         | <b>Data Integration &amp; Missing Data Imputation:</b><br>Consolidate the client profiling metrics with behavioral logs on the matching ID column and fill missing categorical slots with a | High                          | High                         | 1        | yes               |

|   |                                                                                                                                                                                                   |        |      |   |     |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|---|-----|
|   | placeholder value ('Unknown').                                                                                                                                                                    |        |      |   |     |
| 2 | <b>Target Binarization Mapping:</b><br>Compress monthly timeline values (C, X, 0-5) into clear binary classifications where severe defaults are categorized as 0 and clean lines are mapped to 1. | High   | High | 2 | yes |
| 3 | <b>Mitigating Data Class Imbalance:</b><br>Apply class balancing techniques directly into the estimators to prevent the prediction system from blindly outputting "Approved" for every profile.   | High   | High | 3 | yes |
| 4 | <b>Model Benchmarking Exploration:</b><br>Train, fine-tune, and cross-evaluate baseline estimators                                                                                                | Medium | High | 4 | yes |

|   |                                                                                                                                                                                  |        |      |   |     |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|---|-----|
|   | against multi-tree ensembles to determine which framework yields the highest minority-class recall.                                                                              |        |      |   |     |
| 5 | <b>Flask Web Deployment:</b><br>Export the finalized pipeline using a model.pkl binary asset and wrap a dynamic frontend HTML layout with a Flask background web routing engine. | Medium | High | 5 | yes |